Day-4
Python - Operators

What Are Operators in Python?

Operators are **symbols** or **words** that perform **operations** on values or variables.
Just like how we use +, -, ×, and ÷ in Maths — Python uses **operators** to perform **calculations**, **comparisons**, **assignments**, and more.

Skill Course

Categories of Operators

- Arithmetic Operators
- Comparison Operators
- Assignment Operators
- Logical Operators
- Identity Operators
- Membership Operators
- Bitwise Operators (Advanced)

Arithmetic Operators :

- Used for mathematical operations such as addition, subtraction, multiplication, and division.

Arithmetic Operators :

Operator	Meaning	Example	Output
+	Addition	5 + 3	8
-	Subtraction	10 - 4	6
*	Multiplication	4 * 2	8
/	Division	8 / 2	4.0
//	Floor Division	7 // 2	3
%	Modulus (Remainder)	7 % 2	1
**	Exponent (Power)	2 ** 3	8

Assignment **Operators :**

Used to assign and update values in variables.

Examples: =, +=, -=, *=, /=

Assignment **Operators :**

Operator	Example	Meaning
=	x = 5	Assign 5 to x
+=	x += 2	Same as x = x + 2
-=	x -= 3	Same as x = x - 3
*=	x *= 2	Same as x = x * 2
/=	x /= 2	Same as x = x / 2

Comparison Operators :

Used to compare two values.

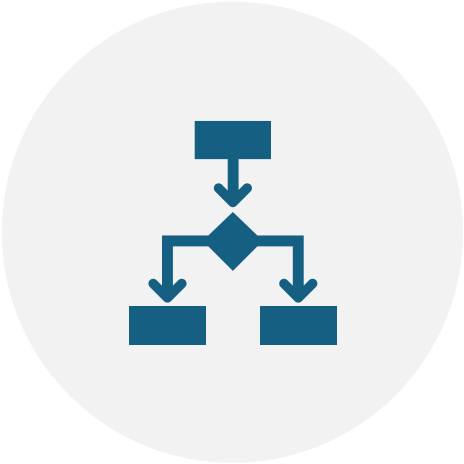
Returns True or False.

Examples: ==, !=, >, <, >=, <=

Comparison Operators :

Operator	Meaning	Example	Result
==	Equal to	5 == 5	True
!=	Not equal to	5 != 3	True
>	Greater than	6 > 3	True
<	Less than	4 < 2	False
>=	Greater or equal	5 >= 5	True
<=	Less or equal	3 <= 2	False

Logical Operators



USED TO COMBINE MULTIPLE CONDITIONS.

EXAMPLES: AND, OR, NOT

Logical Operators

Operator	Meaning	Example	Result
and	True if **both** are True	(5 > 3 and 4 > 2)	True
or	True if **any one** is True	(5 < 3 or 4 > 2)	True
not	Opposite of the condition	not (5 > 3)	False

Identity Operators

Check whether two variables point to the same memory location.

Examples: is, is not

Identity Operator

Operator	Example	Meaning
is	x is y	True if x and y are same object
is not	x is not y	True if x and y are not same object

Membership Operators

Check if a value exists inside a sequence (list, string, tuple).

Examples: in, not in

Membership Operators

OPERATOR	EXAMPLE	RESULT
in	"a" in "apple"	True
not in	"x" not in "apple"	True

Bitwise Operators (Advanced)
Used for **binary operations** (optional for beginners)

Operator	Meaning
&	AND
`	`
^	XOR
~	NOT
<<	Left Shift
>>	Right Shift

🎓 Real Life Example Using Operators:

```
price = 1000
discount = 200
final_price = price - discount
print("Final Price is", final_price)

# Output: Final Price is 800
```